

Generation Interconnection System Impact Study Report

For Queue Project Q14-0337

Old Chapel 138 kV
December 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0337. The Interconnection Customer proposes a 50.0 MW natural gas combined cycle facility interconnecting at the Old Chapel 138 kV substation in the Virginia area. The request was received on June 06, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 183 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,665,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Wheeler / Pasco Cnty Solid Waste Resource Recovery 138kV: 115.6% loading (384 MVA) following the outage of the Arcola / Twin Lakes Alte 138kV; DFAX = 12.5%.

The Opequon / N Mesa 138kV is overloaded to 111.1% of its 263 MVA emergency rating for the loss of the Clrk County / Kearneyville 345kV. The project contributes 7.5% DFAX (3.7 MW) to this overload.

For the loss of Jet / Huntingdon District 138kV, the Greenway / Cleburne 138kV loads to 111.1% of its 177 MVA rating. Project DFAX: 7.6%.

Lcec No. 5 / Opequon 138kV: 110.4% loading (256 MVA) following the outage of the Clark County / Kearneyville 345kV; DFAX = 7.5%.

For the loss of Santa Rosa Energy Center / Balch 138kV, the Creston Nelson / Wpcp 138kV loads to 108.9% of its 228 MVA rating. Project DFAX: 19.9%.

E Ellensburg / Terror Lake Mirogrid 138kV line: 108.7% loading (237 MVA) following the outage of the International Paper Valliant Ok Substation / Kearneyville 345kV; DFAX = 14.9%.

For the loss of Santa Rosa Energy Ctr / Balch 138kV, the Creston Nelson / 3 Mile Knoll 138kV loads to 107.8% of its 222 MVA rating. Project DFAX: 19.9%.

For the loss of Artesia Country Club / Air Force Base 345kV, the Artesia Country Club 138-1 kV Transformer loads to 107.5% of its 465 MVA rating. Project DFAX: 7.0%.

The Artesia Country Club / Ross Fork 138kV is overloaded to 106.4% of its 529 MVA emergency rating for the loss of the Air Force Base / Artesia Country Club 345kV. The project contributes 6.7% DFAX (3.4 MW) to this overload.

The Ocean Rd / Monolith 138kV is overloaded to 106.3% of its 239 MVA emergency rating for the loss of the International Paper Valliant Ok / Kearneyville 345kV. The project contributes 9.2% DFAX (4.6 MW) to this overload.

For the loss of International Paper Valliant Ok / Kearneyville 345kV, the E Ellensburg / W Hills 138kV loads to 106.1% of its 231 MVA rating. Project DFAX: 14.9%.

University Of Maine / Three Peaks 138kV: 104.0% loading (227 MVA) following the outage of the International Paper Valliant Ok / Kearneyville 345kV; DFAX = 9.2%.

The Plum Pt Energy Station / University Of Maine 138kV line is overloaded to 103.4% of its 222 MVA emergency rating for the loss of the International Paper Valliant Ok / Kearneyville 345kV. The project contributes 9.2% DFAX (4.6

MW) to this overload.

For the loss of Kearneyville / International Paper Valliant Ok 345kV, the Ocean Rd / Three Peaks 138kV loads to 103.3% of its 232 MVA rating. Project DFAX: 9.2%.

The Plum Pt Energy Sta / Mabton 138kV is overloaded to 101.5% of its 212 MVA emergency rating for the loss of the International Paper Vlliant Ok Substation / Kearneyville Substation 345kV. The project contributes 9.2% DFAX (4.6 MW) to this overload.

For the loss of International Paper Valliant Ok / Kearneyville 345kV Line, the Southampton / Mabton 138kV loads to 100.6% of its 208 MVA rating. Project DFAX: 9.2%.

Pasco Cnty Solid Waste Resource Recovery / Vmea Peaking Gen 138kV Line: 100.2% loading (219 MVA) following the outage of the Pasco Cnty Solid Waste Resource Recovery / Novec Paradise 138kV; DFAX = 8.5%.

The Southampton Substation / Celeron 138kV is overloaded to 100.0% of its 200 MVA emergency rating for the loss of the International Paper Valliant Ok / Kearneyville 345kV Line. The project contributes 9.2% DFAX (4.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pasco Cnty Solid Waste Resource Recovery / Wheeler 138kV Line	Rebuild line with higher-capacity conductor	\$4,763,000	3% (\$126,000)
Opequon / N Mesa 138kV	Rebuild line with higher-capacity conductor	\$2,730,000	6% (\$171,000)
Greenway / Cleburne 138kV	Rebuild line with higher-capacity conductor	\$1,360,000	3% (\$41,000)
Lcec No. 5 / Opequon 138kV	Rebuild line with higher-capacity conductor	\$6,548,000	6% (\$409,000)
Creston Nelson / Wpcp 138kV	Reconductor line	\$1,085,000	22% (\$233,000)
E Ellensburg / Terror Lake Mirogrid 138kV line	Reconductor line	\$1,546,000	14% (\$212,000)
Creston Nelson / 3 Mile Knoll Substation 138kV L	Reconductor line	\$1,723,000	22% (\$370,000)
Artesia Country Club 138-1 kV Auto	Replace transformer with higher-rated unit	\$4,525,000	6% (\$296,000)
Artesia Country Club / Ross Fork 138kV	Reconductor line	\$1,417,000	4% (\$60,000)
Monolith / Ocean Road 138kV	Reconductor line	\$395,000	6% (\$24,000)
E Ellensburg / West Hills 138kV	Reconductor line	\$1,502,000	14% (\$206,000)
University Of Maine Substation / Three Peaks 138kV	Reconductor line	\$1,276,000	6% (\$76,000)
Plum Point Energy Station / University Of Maine 138kV	Reconductor line	\$308,000	6% (\$18,000)
Three Peaks / Ocean Road 138kV	Reconductor line	\$2,199,000	6% (\$131,000)
Mabton / Plum Point nergy Sta 138kV	Reconductor line	\$1,649,000	6% (\$98,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Southampton / Mabtn 138kV L	Reconductor line	\$1,912,000	6% (\$114,000)
Pasco Cnty Solid Waste Resource Recovery / Vmea Peaking Gen 138kV Line	Reconductor line	\$1,793,000	2% (\$43,000)
Celeron / Southampton 138kV	Reconductor line	\$619,000	6% (\$37,000)
	Total allocated to project		\$2,665,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.